

Multiple Choice Questions

Q1. **Answer:** A

Options: A) Generative models model $P(x, y)$, while discriminative models model $P(y | x)$; B) Discriminative models learn the data distribution and use that knowledge for classification; C) Discriminative models require no training data; D) Generative models are less explainable

Note: Correct option: A - Generative classifiers model the joint distribution $P(x, y)$; discriminative classifiers model the conditional $P(y | x)$.

Q2. **Answer:** B

Options: A) Laplace smoothing; B) Regularization; C) TF-IDF; D) Softmax normalization

Note: Correct option: B - Regularization penalizes large weights (e.g., L1/L2) to reduce overfitting.

Q3. **Answer:** B

Options: A) It cannot use vector representations; B) It only works for linearly separable data; C) It requires probabilistic outputs; D) It cannot be trained with gradient descent

Note: Correct option: B - The perceptron finds a linear decision boundary; it cannot classify non-linearly separable data.

Q4. **Answer:** C

Options: A) It improves cosine similarity; B) It reduces dimensionality; C) It avoids zero probabilities for unseen words; D) It speeds up gradient descent

Note: Correct option: C - Laplace smoothing adds a small count so that unseen words (e.g., in test data) get non-zero probability instead of zero.

Q5. **Answer:** B

Options: A) SGD updates weights using the entire dataset every step; B) SGD updates weights using one or a subset of training examples; C) GD converges faster because updates are noisier; D) SGD cannot be used with neural networks

Note: Correct option: B - SGD uses one or a mini-batch of examples per update; full-batch GD uses the entire dataset every step.

Q6. **Answer:** B

Options: A) Words are represented by their dictionary definitions; B) Words that appear in similar contexts have similar meanings; C) Words are clustered by part of speech; D) Words are ranked by frequency

Note: Correct option: B - The distributional hypothesis: similar context implies similar meaning; embeddings capture this.

Q7. **Answer:** D

Options: A) To penalize large model weights; B) To reduce feature dimensionality; C) To select the class with the largest raw score without normalization; D) To convert raw class scores into a valid probability distribution

Note: Correct option: D - Softmax converts logits (raw scores) into probabilities that sum to 1.

Q8. **Answer:** C

Options: A) After every training example; B) Only when the prediction is correct; C) Only when the prediction is incorrect; D) Only after one full pass over the data

Note: Correct option: C - The perceptron updates weights only on misclassification (when the prediction is incorrect).

Q9. **Answer:** D

Options: A) Increase the weight of very common words; B) Represent word order; C) Convert sparse vectors into dense embeddings; D) Highlight words that are frequent in a document but rare across documents

Note: Correct option: D - TF-IDF upweights terms that are frequent in the document (TF) but rare in the corpus (IDF), capturing discriminative content.

Q10. **Answer:** A

Options: A) It ignores vector length while calculating similarity; B) It increases the influence of high-frequency words; C) It converts similarity scores into probabilities; D) It reduces the dimensionality of word embeddings

Note: Correct option: A - Cosine similarity is based on angle (direction), not magnitude, so vector length does not affect the similarity score.

End of Answer Key